Climate or Design? Mission-Profile Thermal Cycling of Electric Two-Wheeler Motor Controllers in Five Cities

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***Abstract*— Electric two-wheelers are sold mostly in hot countries, and their motor controllers use low-cost discrete MOSFETs. This paper asks how much a hot climate adds to the thermal cycling of such a controller, and how that compares with design and use choices. A 2 kW, 72 V scooter is driven on the WMTC test cycle with the official road load of UN GTR No. 2. Losses of an inverter built from IRFB4115 MOSFETs are taken from the datasheet, and a heatsink model turns them into junction temperature, second by second, for a full year of 2025 NASA POWER weather in Karachi, Delhi, Jakarta, Lagos and Madrid, for a commuter and a delivery rider. Cycles are counted by rainflow and weighted with a Coffin-Manson-Arrhenius law anchored to the part's own qualification test. For cycles caused by riding, the hot cities see 1.1 to 2.6 times the damage of Madrid, and Delhi ranks first under 63 of 72 life-model choices; most of the spread comes from the assumed activation energy. Design and use factors move the damage far more: a poorly cooled box multiplies it by about 17, using 12 instead of 18 MOSFETs by about 9. In absolute terms the cycling damage stays small even in a hard-use case, so hot-climate problems are more likely to appear as thermal-limit events (13.5 riding hours a year above an 80 °C heatsink in Delhi delivery use) than as solder or bond-wire wear-out.**

***Index Terms*—** *electric two-wheeler, motor controller, MOSFET, mission profile, thermal cycling, rainflow, reliability*

I. Introduction

Electric scooters and motorcycles are growing fast in South and Southeast Asia. In Pakistan, for example, about 33,000 electric motorbikes were sold in 2024 and about 57,800 in the first seven months of 2025 [1]. The controller of such a vehicle is a three-phase inverter, usually built from discrete TO-220 MOSFETs on an aluminium case. It works in heat, in stop-and-go traffic, and often with a passenger or cargo.

Mission-profile methods, which turn a real use pattern into temperature swings and then into consumed life, are now a standard reliability tool for power electronics [2]. They have been applied to PV inverters in different climates [3] and to SiC inverters for motor drives, including electric traction [4]. For electric two-wheelers, recent work has looked at the thermal behaviour of the hub motor [5] and at the heatsink of the controller in steady and transient runs [6], but we did not find a study that follows the controller's semiconductors through a year of real weather and compares climates.

One difficulty is that the usual life models (LESIT [7], CIPS08 [8]) were fitted for IGBT modules, and in a recent review the only model fitted on discrete packages comes from TO-247 IGBTs, none from discrete MOSFETs [9]. We therefore use the life law only to rank conditions, check how the ranking depends on the model, and anchor the scale to the qualification test that the manufacturer published for this exact part [10].

The contributions are: (i) an open pipeline in which every input comes from a named public source (GTR No. 2, the MOSFET datasheet, NASA POWER); (ii) a comparison of five cities for two kinds of riders; (iii) a ranking of design and use factors against climate; and (iv) a check of all rankings over 36 life-model settings.

II. Models

A. Vehicle and drive cycle

The vehicle is a scooter of the type sold in Pakistan: 102 kg dry mass, 2 kW hub motor, 70 km/h top speed, 90/90-10 tyres [11], 72 V battery [12]. With the 75 kg driver of GTR No. 2 the mass in running order is $m_{ref} = 177$ kg. The road load is the running resistance table of GTR No. 2, Annex 3 [13]: for $175 < m_{ref} \leq 185$ kg, $F = a + bv^2$ with $a = 15.8$ N and $b = 0.0227$ N/(km/h)^2. For rotating parts we add 7 % of the mass: the regulation's estimates of 3 % for the front wheel and 4 % for the rear wheel with the parts rotating with it, here the hub motor. The table was made for conventional motorcycles and is used here as the official reference.

The speed trace is part 1 of the WMTC (600 s, 4.065 km, up to 60 km/h), read from the tables of GTR No. 2 [13]. By maximum speed the scooter falls in the 50–100 km/h band of subclass 1-3, which uses part 1; an electric vehicle has no engine capacity, so this mapping is our choice.

WMTC asks for up to about 4.1 kW, more than a 2 kW scooter delivers, so the controller limits the motor to a 100 A peak phase current and 3 kW from the battery, of which 88 % reaches the wheel (assumed values, varied later). The scooter follows the trace as closely as these limits allow; the RMS speed error is 1.6 km/h. The model needs 18.7 Wh/km at the wheel.

The claimed 120 km range on a 2.52 kWh pack implies about 21 Wh/km at the battery, which leaves room for motor and controller losses, so we take the model as plausible.

The motor is a direct-drive PMSM under field-oriented control. Its back-EMF constant is set so that the phase back-EMF at top speed is 90 % of $V_{dc}/\sqrt{3}$, giving $k_e = 0.40$ V s/rad and a peak phase current $I = T/(1.5k_e)$. There is no regenerative braking in the base case.

B. MOSFET losses

Each of the six switches has N IRFB4115 devices in parallel, $N = 3$ in the base case (an "18-MOSFET" controller). From the datasheet [14]: $R_{DS(on)} = 9.3$ mΩ typical at 25 °C with the temperature curve of its Fig. 4 (read by eye, about ±0.05), $t_r = 73$ ns, $t_f = 39$ ns, $Q_{rr} = 300$–450 nC. With complementary gating each switch carries on average $P_{cond} = R_{DS(on)}(T_J)\,(I/N)^2/4$. Switching loss is linear in current and voltage, $E = \frac{1}{2}V_{dc}\bar{I}(t_r + t_f)$ with $\bar{I} = 2I/(\pi N)$, plus $Q_{rr}V_{dc}$, at 15 kHz. Because a real gate drive is slower than the datasheet test circuit, the switching times are doubled in the base case. At 100 A and $T_J = 50$ °C the loss is 4.6 W per device, two thirds of it conduction.

C. Thermal model

The datasheet gives $R_{thJC} = 0.4$ K/W as a two-element Foster network with time constants of 5.9 and 0.63 ms, and $R_{thCS} = 0.5$ K/W. At 1 s steps the junction-to-case drop is therefore quasi-static, $T_J = T_{hs} + P(R_{thJC} + R_{thCS})$. The controller box is one thermal mass, $C_{hs}\dot{T}_{hs} = P_{tot} - (T_{hs} - T_{amb})/R_{hs}$, with $C_{hs} = 450$ J/K (about 0.5 kg of aluminium) and $R_{hs} = 0.5$ K/W while riding. Both are assumptions and are varied.

D. Weather and use

T_{amb} is the hourly 2 m air temperature (MERRA-2 based) from NASA POWER [15] for 2025 at the five city centres. In earlier work on the Karachi files we found that POWER's "LST" is the nominal zone of the longitude, not civil time [16]; we apply the same rule to all cities, with summer time for Madrid. Table 1 gives the climates: Delhi has the hottest hours (956 above 35 °C, maximum 45.9 °C), Jakarta and Lagos are warm but never reach 35 °C, and Madrid has a cool mean but 212 hours above 35 °C.

Two riders are simulated. The commuter rides Monday to Saturday at 08:00 and 17:30, each ride three WMTC part 1 cycles (30 min, 12 km). The delivery rider rides every day, twelve rides from 11:00 to 22:00, each two cycles (20 min, 8 km), about 96 km a day. All 365 days are stepped together, one second at a time.

E. Cycle counting and damage index

The junction temperature is reduced to reversals with 0.5 K hysteresis and counted with the three-point rainflow method of ASTM E1049 [17], which we implemented and tested against a hand count. Two countings are compared: per day (all cycles, including the slow swing of the air temperature between a morning and an evening ride) and per ride (only the cycles caused by riding, each ride plus 15 min of cooling).

Table 1. Climate and base-case results (delivery rider; damage per year)

	Karachi	Delhi	Jakarta	Lagos	Madrid
mean T_{amb} (°C)	26.3	24.8	27.6	26.8	14.4
max T_{amb} (°C)	40.4	45.9	32.0	32.2	40.6
hours > 35°C	173	956	0	0	212
max T_J (°C)	52.5	58.2	43.4	43.7	52.6
D, per-ride count (10^{-7})	6.9	8.2	6.3	6.0	3.7
ratio to Madrid	1.86	2.22	1.71	1.61	1.00

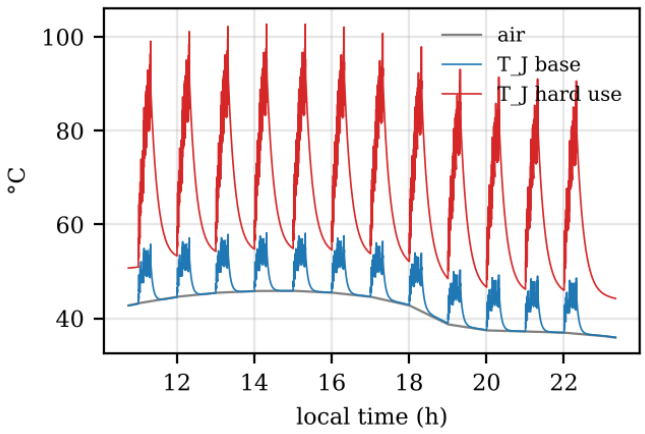


Figure 1. Delivery rider on the hottest day of 2025 in Delhi: air temperature, junction temperature in the base case, and in the hard-use case of Section III-D (whose air is 8 K warmer inside the body).

Damage is $D = \sum n_i/N_f(\Delta T_i, T_{m,i})$ with $N_f = A\,\Delta T^{-n}\exp(E_a/k_B T_m)$. The qualification report of IRFB4115 lists an intermittent operational life test (MIL-STD-750 method 1037) at $\Delta T = 100$ K for 8572 cycles with no failure in 231 parts [10]. We set $N_f(100\text{ K}) = 8572$ with an assumed mean of 75 °C. This treats a passed test as a failure, so D is pessimistic; $D = 1$ means as much cycling as the whole test, not end of life. The base model uses $n = 5$ from the LESIT model and $E_a = 0.617$ eV from a Coffin-Manson-Arrhenius module model, both as listed in [9]; $n = 3$–6, $E_a = 0$–0.617 eV and a minimum counted range of 0, 2 or 5 K give 36 settings. Ratios between cities do not depend on A.

III. Results

A. Base case

The controller runs cool in the base case (Table 1). The mean junction rise over the air while riding is 5.9–6.6 K in every city, and the hottest junction of the year is 41–58 °C. The cities differ in the level the cycles ride on, not in their size. Fig. 1 shows the delivery rider on Delhi's hottest day.

B. How much does the climate matter?

Counting only the cycles caused by riding (Fig. 2a), every hot city sees more damage than Madrid: 1.6 to 2.2 times for the delivery rider and 2.0 to 2.4 times for the commuter in the base model, and 1.1 to 2.6 times over all 36 settings. Delhi is the most damaged city in 35 of 36 settings for the delivery rider

and in 28 of 36 for the commuter (Jakarta in the others). Most of the effect comes from the activation energy: with $E_a = 0$ the ratios fall to 1.13–1.50, which is what the larger $R_{DS(on)}$ at high temperature and the larger cycles alone give.

The whole-day counting tells a different story for the commuter (Fig. 2b): Madrid becomes the most damaged city in 30 of 36 settings. In Madrid 97 % of the commuter's damage comes from half cycles, the dry-climate swing of the air (on average 11.9 K between the coolest and warmest moment of the riding day, against 2.4 K in Jakarta). Power-cycling laws were fitted to cycles of seconds to minutes, not to swings of ten hours, so we regard the per-ride result as the more reliable one and report the other as a warning: the choice of counting can reverse a climate ranking.

C. Design and use against climate

Fig. 3 changes one factor at a time for the delivery rider in Delhi (per-day counting). A box with no airflow ($R_{hs} = 1.5$ K/W) multiplies the damage by 17.0, a 12-MOSFET controller by 9.2, a controller tuned for 150 A and 4.5 kW by 5.8, a gate drive four times slower than the datasheet by 3.9, and a passenger or 75 kg of cargo by 2.8. Eight kelvin of extra heat inside the body, which is a climate-like change, gives 2.1, regenerative braking 1.8 and the worst-case $R_{DS(on)}$ 1.6. The same order holds in Karachi and Madrid. Every design and use factor we tried is of the same size as the climate effect or larger.

D. Hard use

The hard-use case combines a box with no airflow, 12 MOSFETs, 8 K of body heat and 75 kg of extra load. The junction then reaches 84–106 °C and the largest cycle of the year for the delivery rider in Delhi reaches 63 K (Fig. 1). With delivery use the heatsink is above 80 °C for 13.5 riding hours a year in Delhi, 1.0 in Madrid, 0.5 in Karachi and never in Jakarta or Lagos. A controller that protects itself by reducing power near such a temperature would do so mostly in Delhi, almost all of it in April, May and June. The damage index rises about 2200 times for the delivery rider in Delhi (between about 1000 and 7000 times across all cases) but stays at 3.5×10^{-3} per year (2.3×10^{-2} with $n = 3$), that is, at most about 2.3 % of the qualification test's cycling each year even with the pessimistic anchor.

IV. Discussion and Limitations

Within this model, climate changes the thermal-cycling damage of a scooter controller by a factor of about two, while cooling and the number of MOSFETs change it by about an order of magnitude. Because the absolute damage is small, a hot climate is more likely to show up as reduced power at the thermal limit, or through mechanisms not modelled here, than as bond-wire or solder fatigue of the MOSFETs. For manufacturers selling into hot cities, the cooling path of the box and the device count are the levers that matter, and summer peaks in places like Delhi are the cases to check.

The study has clear limits. Nothing is validated against measurements of a real controller; the box thermal resistance and capacity, the controller limits, the body-heat offset and the use profiles are assumptions, which is why they are varied. The road load table was written for conventional motorcycles, WMTC has no hills, and real city traffic in these countries may be harsher than WMTC. Losses are averaged over an electrical period, which misses the slow current pulsation at walking speed. The air temperature is not the temperature inside the scooter in the sun. The life law is extrapolated from 100 K to cycles of a few kelvin and was fitted for modules, so the absolute damage index is an indicator, not a lifetime. Only one year of weather is used.

V. Conclusion

An open, fully sourced mission-profile model of an electric two-wheeler controller shows that hot cities add roughly 1.1–2.6 times the thermal cycling of Madrid, with Delhi worst in 63 of 72 settings, while design and use factors change it by up to 16–21 times. The result also depends on how the cycles are counted, which future studies should state. Measuring the case temperature of a real controller in summer delivery use would be the next step, and would let the assumed thermal parameters be replaced by data.

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Code and Data

`https://github.com/tehreem-gillani/e2w-controller-wearout`. Weather: NASA POWER hourly 2025. Drive cycle and road load: UN GTR No. 2. Device data: Infineon IRFB4115PbF datasheet and product qualification report.

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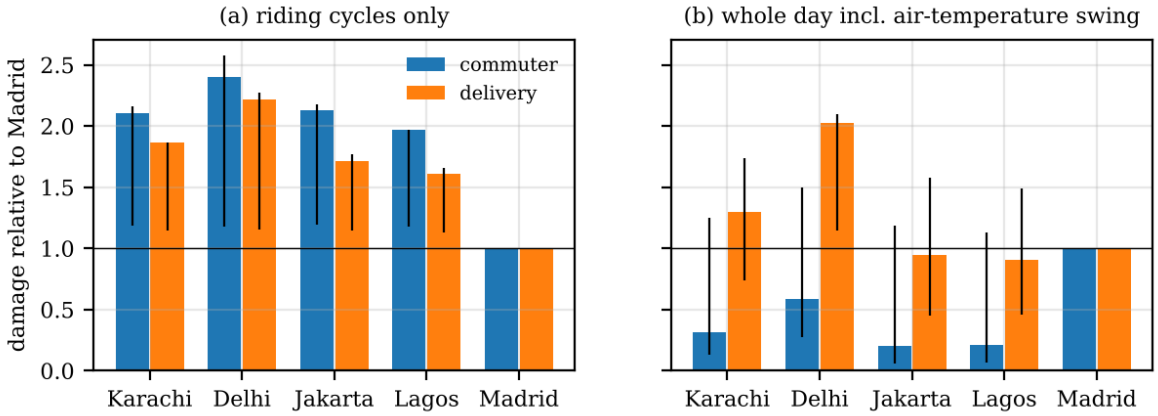


Figure 2. Damage relative to Madrid. Bars: base life model ($n = 5$, $E_a = 0.617$ eV, ranges ≥ 2 K). Lines: range over all 36 life-model settings. (a) cycles counted per ride; (b) counted per day, including the slow swing of the air temperature.

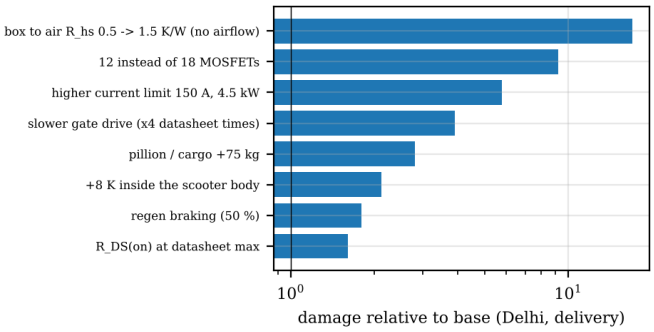


Figure 3. One factor at a time, delivery rider in Delhi: damage relative to the base case (log scale).

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